St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

Question Bank (Revised Curriculum 2018 onwards)

First Year- Second Semester (2019 -21 Batch)

Course Title (Paper Title): SEPARATION TECHNIQUES QP Code: 55101

Unit	SI. No	Questions	Marks
1	1	Write Van Deemter's equation and explain the terms.	2
1	2	Define frontal and displacement analysis.	2
1	3	How chromotographic separation is optimized using capacity factor?	2
1	4	How chromotographic separation is optimized using selectivity factor?	2
1	5	How chromotographic separation is optimized using column efficiency?	2
1	6	Why guard column is used before analytical column in HPLC?	2
1	7	Write a Van Deemter equation.	2
1	8	Explain the terms : i) Mobile Phase ii) Stationary Phase	2
1	9	Write the VAN Deemter equation and explain the terms in it.	2
1	10	Describe the criteria used in the selection of stationary phases in chromatography.	2
1	11	Write the applications of column chromatography with reference to pharmaceutical chemistry.	2
1	12	How are the chromatographic techniques classified based on the physical properties?	2
1	13	Write the Golay equation and define the terms in it.	2
1	14	What do you mean by capacity factor?	2
1	15	What is chromatographic resolution?	2
1	16	What is retention time and Rf value?	2
1	17	Give the principles of HPLC & GLC.	2
1	18	What are the characteristics of an ideal detector for HPCL?	2
1	19	Why ECD is superior to FID and TCD in gas chromatography?	2

1	**20**	Differentiate between isocratic and gradient elusion.	2
1	**21**	What is retention time and Rf value? Give their significance	2
1	**22**	Define the following terms :1. Frontal analysis 2. Displacement analysis	2
1	**23**	Explain distribution co-efficient.	2
1	**24**	Distinguish between the terms distribution co-efficient and distribution ratio.	2
1	**25**	Compare classical column chromatography with HPLC.	2
1	**26**	Distinguish packed bed chromatography and open tabular (capillary) chromatography.	2
1	**27**	What is retention volume?	2
1	**28**	What is baseline width?	2
1	**29**	Write Van Deemter Equation.	2
1	**30**	Write Golay Equation.	2
1	**31**	Write Huber-Knox Equation.	2
1	**32**	Give the principle of Gas chromatography.	2
1	**33**	Give the principle of HPLC.	2
1	**34**	List out the qualitative applications of GC.	2
1	**35**	Mention the quantitative applications of GC.	2
1	**36**	Mention the qualitative applications of HPLC.	2
1	**37**	List out the quantitative applications of HPLC.	2
1	**38**	Explain peak capacity.	2
1	**39**	Explain stationary phase and mobile phase	2
1	**40**	In a chromatographic analysis of lemon oil a peak for limonene has a retention time of 8.36 min with a baseline width of 0.96 min. γ-terpinene elutes at 9.54 min. with a baseline width of 0.64 min. What is resolution between the two peaks?	2
1	**41**	Give the relationship between selectivity factor and their retention factor	2

1	**42**	How chromotographic separation is optimized using column selectivity?	2
1	**43**	Explain non ideal behavior of chromatography	2
1	**44**	Give the applications of column chromatography.	2
1	**45**	Mention the applications of GC.	2
1	**46**	List out the applications of HPLC.	2
1	**47**	What is isocratic and gradient elusion?	2
1	**48**	Give the characteristic features of HPLC.	2
1	**49**	What is distribution constant?	2
1	**50**	What is normal and reverse phase mode?	2
1	**51**	Mention the characteristic features of HPLC	2
1	**52**	Compare HPLC with GLC.	2
1	**53**	Compare HPLC with classical column chromatography .	2
1	**54**	Define distribution constant	2
1	**55**	Differentiate isocratic and gradient elusion.	2
1	**56**	Differentiate between normal and reverse phase mode	2
1	**57**	Define displacement chromatography.	2
1	**58**	Write the relationship between selectivity factor and their retention factor.	2
2	**59**	What will be the composition of the effluent when a dilute solution of each of the following is passed through cation exchange column in the hydrogen form (a) NaCl; (b) Na_2SO_4; (c) $HClO_4$; (d) $FeSO_4$; (e) $(NH_4)_2SO_4$ f) $MgSO_4$	02
2	**60**	Explain the difference between cation exchange resin and an anionic exchange resin	02
2	**61**	Write the structures of polystyrene and polydivinylbenzene for the use of cationic and anionic exchange resin columns.	02
2	**62**	How are ion exchangers classified broadly and on the basis of category of	02

		material ? Give a suitable example for each type	
2	**63**	What are the different types of synthetic inorganic ion exchangers ? Discuss their special properties.	02
2	**64**	Explain the difference between adsorption and ion-exchange processes.	02
2	**65**	What is the basic difference between Size exclusion and ion exchange?	02
2	**66**	What are the main advantages of majority of inorganic ion exchangers over their organic counterparts?	02
2	**67**	Name the factors on which the selectivity of an ion exchanger for an ion depends	02
2	**68**	What are the important characteristics of a useful ion exchanger?	02
2	**69**	How does ion exchange help in improving the quality of alcoholic drinks?	02
2	**70**	Describe the fundamental difference between ion-exchange and size exclusion chromatography	02
2	**71**	Show that the distribution co-efficient in IEC is pH dependent	02
2	**72**	A sodium phosphate solution is passed through a column of an anion exchanger in the chloride form. The $PO4^{3-}$ ions are taken up by the ion exchanger. Write down the ion exchange equilibria.	02
2	**73**	Describe the differences between single column and suppressor-column ion chromatography	02
2	**74**	What are the important characteristics of a useful ion exchanger ?	02
2	**75**	What is the role of divinylbenzene in the synthesis of styrene-divinylbenzene polymeric ion exchange resins ?	02
2	**76**	What is the basic difference between adsorption and ion exchange?	02
2	**77**	A sodium sulphate solution is passed through a column of an anion exchanger in the chloride form. The sulphate ions are taken up by the ion exchanger. Write down the ion exchange equilibria.	02
2	**78**	Describe the fundamental difference between ion-exchange and size exclusion chromatography	02

2	**79**	Describe the principle of SEC. what is exclusion limit.	02
2	**80**	Among DNA and protein which will be eluted first during the desorption process assuming that both are of similar molecular mass. Justify your answer with suitable explanation.	02
2	**81**	Illustrate the synthesis, properties and applications of Dextran Gels.	02
2	**82**	Draw the structure of Silica gel depicting the various types of hydroxyl groups. What is the order of increasing activity ? How are siloxanes prepared ?	02
2	**83**	Explain the term 'Retention Time' with the help of a plot. What are the consequences when the retention factor is (i) too small, and (ii) too large ?	02
2	**84**	What is the basic equation for size exclusion chromatography? Under what conditions the value of K, used in the equation, exceeds unity?	02
2	**85**	Mention the characteristics which are important for a gel to be useful for chromatographic work.	02
2	**86**	How is gel filtration chromatography used for molecular weight determination of proteins?	02
2	**87**	What particular property of gel is responsible for fractionation of solutes of different molecular weights by size exclusion chromatography?	02
2	**88**	What are the values of distribution coefficient if a)solute molecules do not enter the gel matrix b) solute molecules enter the gel matrix?	02
2	**89**	Mention three unique advantages of size exclusion chromatography.	02
2	**90**	What is retention time and R_f value?	02
2	**91**	What are the differences between paper chromatography and thin layer chromatography ?	02
2	**92**	Explain Rf value and various factors affecting it.	02
2	**93**	Describe the effect of pressure on supercritical fluid chromatograms	02
2	**94**	List some advantageous properties of supercritical carbon di oxide as a mobile phase for Chromatographic separations	02

2	**95**	Define (a) critical temperature and critical pressure of a gas (b) supercritical fluid	02
2	**96**	State the advantages of Supercritical Chromatography.	02
2	**97**	How will you separate and identify Mn2+, Cu2+ and Zn2+ in a mixture by paper chromatography ? In a separation procedure, solvent went up to 7.6 cm, whereas spots were observed at 2.3, 5.8 and 6.5 cm, respectively. Calculate their Rf values.	02
2	**98**	In a paper chromatographic separation of cations- Ag^{+}, Pb^{+} and Hg^{+}, solvent front rises to 18.4 cm while cationic spots were observed at 15.8, 12.1 and 5.9 cm, respectively. Calculate Rf values of the metal ions. Emission gases of a two wheeler were tested for pollutant metals by paper chromatography. A spot corresponding to Rf value of 0.65 was observed. What is the possible pollutant metal ion in the emission gases?	02
2	**99**	Describe the preparation of exactly 2.00 L of 0.1500 M HCl from primary standard grade NaCl using a cation exchange resin.	02
3	**100**	What is electro osmotic flow? Why does it occur?	02
3	**101**	Suggest how electro osmotic flow might be suppressed	02
3	**102**	Describe the basis of separation in electrophoresis	02
3	**103**	Describe the principles of capillary electrophoresis. What are its advantages?	02
3	**104**	Write the principle of capillary electrophoresis.	02
3	**105**	Why does pH affect separation of amino acids by electrophoresis?	02
3	**106**	Three large proteins are ionised at the pH at which an electrical separations is carried out. If the ions are designated A^{2+}, B^{+}, and C^{3+}, predict the order of elution.	02
3	**107**	Explain the terms osmotic mobility and electrophoretic mobility and give their equations	02
3	**108**	Why does the internal wall of the capillary tube attract positively charged ions?	02
3	**109**	What is the importance of loading buffer in DNA gel electrophoresis?	02

3	**110**	How are samples introduced into the capillary electrophoresis?	02
3	**111**	What is the role of ethidium bromide in DNA gel electrophoresis?	02
3	**112**	Suggest a method for the separation of aniline, an organic base from nitrobenzene.	02
3	**113**	Suggest a method for the separation of aniline from benzoic acid	02
3	**114**	Suggest a method for the separation of phenol from aniline	02
3	**115**	Suggest a method for the separation of acetic acid from aniline	02
3	**116**	Define is Nernst distribution coefficient with suitable examples	02
3	**117**	Explain distribution law and its limitations in solvent extraction process.	02
3	**118**	What is synergic extraction? Indicate its benefits in the extraction of metals as neutral chelates	02
3	**119**	Distinguish between the terms distribution co-efficient and distribution ratio.	02
3	**120**	Describe the equilibrium process involved in the solvent extraction of metal chelates.	02
3	**121**	How will you classify the solvent extraction methods based on equilibrium processes ?	02
3	**122**	While proposing the metal chelate extraction equilibria, two simplifications with regard to the existence of the metal in the aqueous and organic phase were assumed.What are they?	02
3	**123**	The values of which constants should be higher for a higher metal chelate extraction?	02
3	**124**	Explain the fundamental difference between continuous and batch extraction	02
3	**125**	Discuss the applications of solvent extraction	02
3	**126**	Discuss the influence of salting out and masking agents in the extraction process.	02
3	**127**	What is the main role of a masking agent in the extraction equilibria of a metal chelate?	02

3	**128**	How can the $pH_{1/2}$ of extraction of a metal chelate be altered?	02
3	**129**	What is solid-phase micro extraction?	02
3	**130**	In what order would be the following compounds are eluted from an alumina column using n-hexane as the eluting solvent? Ethyl alcohol, acetaldehyde, acetic acid.	02
3	**131**	A Method describes the analysis of the trihalomethanes $CHCl_3$, $CHBr_3$, $CHCl_2Br$, and $CHClBr_2$ in drinking water using a packed column with a nonpolar stationary phase. Predict the order in which these four trihalomethanes will elute.	02
3	**132**	A mixture of n-heptane, tetrahydrofuran, 2-butanone and n-propanol elutes in this order when using a polar stationary phase such as carbowax. The elution order is exactly the opposite when using a non-polar stationary phase such as polydimethyl siloxane. Explain the order of elution in each case.	02
3	**133**	Bring out the difference between the solid phase extraction and solvent extraction.	02
3	**134**	Ninety six percent of a solute is removed from 100 ml of an aqueous solution by extraction with 50 ml portions of an organic solvent. What is the distribution ratio of the solute?	02
3	**135**	The distribution ratio between 3 M HCl and tri-n-butylphosphate for $PdCl_2$ is 2.3. What percent $PdCl_2$ will be extracted from 25.0 ml of a 7.0×10^{-4} M solution into 10.0 ml tri-butylphosphate	02
3	**136**	Ninety percent of a metal chelate is extracted when equal volumes of aqueous and organic phases are used. What will be percent extracted if the volume of organic phase is doubled?	02
3	**137**	Ninety percent of a metal chelate is extracted when equal volumes of aqueous and organic phases are used. What will be percent extracted if the volume of organic phase is tripled?	02
3	**138**	Arsenic (III) is 70% extracted from 7 M HCl into an equal volume of toluene. What percent will remain unextracted after three individual extractions with toluene?	02

3	**139**	Metal is 70% extracted from 7 M HCl into an equal volume of toluene. What percent will remain unextracted after four individual extractions with toluene?	02
3	**140**	Twenty millilitres of an aqueous solution of 0.10 M butyric acid is shaken with 10 ml ether. After the layers are separated, it is determined by titration that 0.5 milli mol butyric acid remains in the aqueous layer. What is the distribution ratio and what is the percent extracted.	02
3	**141**	Eighty percent of a metal chelate is extracted when equal volumes of aqueous and organic phases are used. What will be percent extracted if the volume of organic phase is tripled?	02
3	**142**	Arsenic (III) is 50% extracted from 7 M HCl into an equal volume of toluene. What percent will remain unextracted after three individual extractions with toluene?	02
3	**143**	Twenty millilitres of an aqueous solution of 0.10 M butyric acid is shaken with 10 ml ether. After the layers are separated, it is determined by titration that 0.35 milli mol butyric acid remaining the aqueous layer. What is the distribution ratio, and what is the percent extracted.	02
3	**144**	The distribution ratio between 3 M HCl and tri-n-butylphosphate for $PdCl_2$ is 10. What percent $PdCl_2$ will be extracted from 25.0 ml of a 7.0×10^{-4} M solution into 10.0 ml tri-butylphosphate	02
3	**145**	**For a representative 50 cm capillary column operated at 30,000 V, the typical net mobility for a 10 min migration time is about 2×10^{-8} m^2 $V^{-1}S^{-1}$. Calculate the number of plates for a small Li^+ ($D=1.0\times10^{-9}$ m^2 s^{-1}) and for a 100,000 MW protein with $D=3 \times 10^{-11}$ m^2 s^{-1}.**	02
1	**146**	List out the criteria for the selection of mobile and stationary phase in HPLC.	05
1	**147**	Give the qualitative and quantitative applications of column chromatography.	05
1	**148**	How chromotographic separation is optimized using capacity factor and column selectivity?	05
1	**149**	With neat diagram explain the working of FID in GC.	05
1	**150**	Write a note on detectors used in HPLC and Applications of HPLC.	05
1	**151**	Write the principles and explain the instrumentation of Gas chromatography.	05
1	**152**	Explain the following terms : 1) Frontal analysis 2) Displacement analysis.	05

1	**153**	Explain frontal and displacement analysis.	05
1	**154**	How chromotographic separation is optimized using capacity factor?	05
1	**155**	Write a note on instrumentation and applications of HPLC.	05
1	**156**	Discuss the working of thermal conductivity detector used in GC.	05
1	**157**	Write a note on factors governing column efficiency in GC.	05
1	**158**	Discuss briefly about FID used in GC.	05
1	**159**	Write a note on detector used in HPLC.	05
1	**160**	Define and explain the terms retention time and capacity factor.	05
1	**161**	What is column efficiency of a chromatographic column? Briefly describe the rate theory and plate theory approach to explain the parameters.	05
1	**162**	Explain the basic instrumentation of a typical HPLC unit for the separation of components.	05
1	**163**	Describe the working principle and advantages of Flame ionisation detector in gas chromatography with a neat labelled diagram.	05
1	**164**	List out the characteristics of an ideal detector used in gas chromatography.	05
1	**165**	Describe the various types of pumps used in HPLC. What are advantages and disadvantages of each?	05
1	**166**	With a neat sketch explain the HPLC instrumentation and the role of each component in it.	05
1	**167**	Describe the working principle of electron capture detector used in GC with neat sketch.	05
1	**168**	List the variables that lead to band broadening in chromatography.	05
1	**169**	What are the major types of detectors used in HPLC? Describe the advantages and disadvantages	05
1	**170**	Discuss the principles of HPLC & GLC.	05
1	**171**	Explain the working of FD detector used in GC with neat diagram.	05
1	**172**	Give the applications of column chromatography in qualitative analysis.	05
1	**173**	Discuss on the types of columns used in GC.	05
1	**174**	Describe plate and rate theory regarding the rate of migration of solute and the development of peaks in chromatogram.	05
1	**175**	Explain the terms : i) Chromatographic resolution ii) Capacity factor iii) Column selectivity iv) Column efficiency	05
1	**176**	Define and explain the terms retention time and retention volume.	05
1	**177**	Explain in detail the classification of chromatography based on the principle of separation.	05
1	**178**	Describe the working principle and advantages of Flame ionization detector in gas chromatography with a neat labeled diagram.	05

1	**179**	What is Van Deemter equation? Define the terms in it. How does the Golay equation differ from Van Deemter equation.	05
1	**180**	Point out the advantages and disadvantages of HPLC over conventional column chromatography	05
1	**181**	Discuss the classification of chromatographic techniques.	05
1	**182**	What is column resolution? Briefly discuss the influence of selectivity factors and retention factor on resolution with help of a typital chromatogram.	05
1	**183**	Draw a neat diagram of GC instrument and describe the function of each component.	05
1	**184**	With a neat diagram explain the working principles of HPLC.	05
1	**185**	Describe the principles upon which flame ionization detector is based. What are its advantages and limitations.	05
1	**186**	Explain briefly the procedure for column packing in HPLC.	05
1	**187**	Explain the factors responsible for band broadening.	05
1	**188**	What is column efficiency of a chromatographic column? Briefly describe the rate theory and plate theory approach to explain the parameters.	05
1	**189**	Draw the schematic diagram of HPLC unit, label the components and explain the function of each component.	05
1	**190**	What do you understand by the terms retention time and retention volume? Write the expression with relates both terms.	05
1	**191**	Explain any two factors influencing band broadening in elution chromatography.	05
1	**192**	Explain in detail the classification of chromatography based on the principle of separation.	05
1	**193**	Explain displacement chromatography.	05
1	**194**	Explain flame ionisation detector used in GC. Mention its advantages.	05
1	**195**	Write the applications of Gas chromatography in qualitative analysis.	05
1	**196**	How chromotographic separation is optimized using column efficiency?	05
1	**197**	How chromotographic separation is optimized using capacity factor?	05
1	**198**	Write a note on non ideal behavior of chromatography	05
1	**199**	Elaborate the principle and instrumentation of GC.	05
1	**200**	Give the principle and instrumentation of HPLC.	05
1	**201**	Write a note on general description of column chromatography.	05
1	**202**	Give the qualitative and quantitative applications of GC.	05
1	**203**	Give the qualitative and quantitative applications of HPLC.	05
1	**204**	Write a note on detectors used in HPLC.	05
1	**205**	Write a note on detectors used in GC.	05

1	**206**	With a neat diagram explain the working principles of column chromatography.	05
1	**207**	Explain the working principles of column chromatography with labeled diagram.	05
1	**208**	Elaborate the working principles of column chromatography with labeled sketch.	05
1	**209**	Explain argon ionisation detector used in GC. Give its advantages and disadvantages.	05
1	**210**	Discuss electron capture detector used in GC. Give its advantages	05
1	**211**	Give the characteristic features of HPLC.	05
1	212	Discuss argon ionisation detector used in GC. Give its advantages and disadvantages.	05
2	**213**	Describe the criteria used in the selection of stationary phases in ion exchange chromatography. Explain the basic features of ion exchange mechanism. How is the ion exchangers classified ? Give an example for each type.	05
2	**214**	Describe the underlying principles of separation of metals ions by ion exchange chromatography.	05
2	**215**	Explain the principle of Ion Exchange Chromatography. Describe the factors that affect the selectivity of ion exchange resin.	05
2	**216**	Show that the distribution co-efficient in IEC is pH dependent. Explain briefly application of ion exchange chromatography.	05
2	**217**	Give a classification of synthetic inorganic ion exchangers. Cite one example of each category. Discuss insoluble acidic salts of polyvalent metals as ion exchangers in detail. What are the general advantages of synthetic inorganic ion exchangers over organic resinous ion exchangers ? Cite one example of application of synthetic inorganic ion exchanger particularly based on each of the advantages	05
2	**218**	Explain the principle involved in separation of high molecular weight compounds by SEC. Discuss the application of size exclusion chromatography.	05
2	**219**	Briefly explain the inherent factors for the selection of stationary phase in size exclusion chromatography.	05
2	**220**	Give an account of principles and methodology of gel permeation chromatography. List their advantages.	05

2	**221**	Explain the basic principle of size exclusion chromatography with the help of an illustration. Draw the nature of plot between log mol wt and retention volume.	05
2	**222**	Discuss the different properties of gels which have to be considered for them to be useful for size exclusion chromatography. How is gel filtration chromatography used for the determination of molecular weight of proteins ?	05
2	**223**	Give a broad classification of different types of gels used for size exclusion chromatography. Discuss the preparation, special features and advantages of Dextran gels (Sephardic).	05
2	**224**	What is Size Exclusion Chromatography ? Discuss the important properties of gels.	05
2	**225**	Describe the principle of size exclusion chromatography in the separation of biomolecules. What is molecular sieving process?	05
2	**226**	Describe the principle of size exclusion chromatography. What is the exclusion limit and inclusion limit	05
2	**227**	Explain the detection methods adopted in Thin layer chromatography.	05
2	**228**	The R_f values in TLC are difficult to reproduce. Explain the factors responsible for the non-reproducibility of Rf values.	05
2	**229**	How a greater separating power can be achieved by two dimensional TLC?	05
2	**230**	Write briefly on the preparation of plates, selection of adsorbents and solvents, identification of spots, and calculation of Rf value in TLC.	05
2	**231**	How are colorless spots identified in TLC. Write in brief about the high performance thin layer chromatography .	05
2	**232**	Explain the preparation of plates in thin layer chromatography. Illustrate the application of TLC in the separation of products.	05
2	**233**	Comment on the quantitative aspects of thin layer chromatography. Explain the different approaches adopted for this purpose. Which of the following types of paper can be used in paper chromatography ? Give reason in support of your answer.(i) Butter paper (ii) Cellophane paper (iii) Cellulose filter paper (iv) Ordinary filter paper (v) Writing paper	05

2	**234**	Describe the various steps involved in thin layer chromatography. Explain the selection of adsorbents and solvents in TLC	05
2	**235**	Discuss the mechanism of separation in affinity chromatography. Explain the role of spacer arms and activating agents in this technique. Mention the type of compounds separated by affinity chromatography.	05
2	**236**	Write the mechanism of separation achieved in affinity chromatography. Give one application of the technique.	05
2	**237**	What is matrix activation in affinity chromatography? Explain the method of matrix activation applied to affinity chromatography	05
2	**238**	Make an account on the steps involved in affinity chromatography. Explain the role of spacer arm and discuss how matrix can be activated to facilitate the coupling of specific types of ligands.	05
2	**239**	Explain the principle involved in the separation of bio components by Affinity chromatography. Discuss the role of spacer arms.	05
2	**240**	Discuss the methods of separation in which partition and surface activity play main role in the separation of the biocomponents of a mixture	05
2	**241**	How do the instruments for supercritical fluid chromatography differ from those for a) HPLC and b) GC. What important property of supercritical fluid is related to their high densities?	05
2	**242**	Compare SFC with other chromatographic methods. List the advantages and disadvantages of SFE compared to liquid-liquid extractions	05
2	**243**	How are analytes usually recovered after an SFE. Differentiate between on-line and off-line extractions	05
2	**244**	Describe the instrumentation and operating variables of supercrititical fluid chromatography. Why are highly polar or ionic compounds usually not eluted in SFC?	05
2	**245**	Explain the advantageous features of supercritical chromatography over other column methods. What is its application in the separation methodology?	05
2	**246**	What properties of a supercritical fluid are important in chromatography? List some advantageous properties of supercritical carbon-di-oxide as a mobile phase for Chromatographic separations.	05

2	**247**	A mixture of U, Mg and Al was separated by paper chromatography and then quantitatively determined by spectrophotometry. 100 ppm standards of each of these were prepared and reacted with oxine for U, magneson for Mg, aluminon for Al giving absorbance values 0.85, 0. 80, 0.65, respectively. Spot areas of paper chromatogram were cut, dissolved and their absorbances were found to be 0.39 for U, 0.42 for Mg and 0.53 for Al. Calculate concentrations of each of three elements in the mixture.	05
2	**248**	The sodium ion in 200 ml of a solution containing 10 g/L is to be removed by passing through a cation exchange column in the hydrogen form. If the exchange capacity of the resin is 5.1 meq/g of dry resin. What is the minimum weight of dry resin required?	05
2	**249**	Alkali metal ions can be determined volumetrically by passing a solution of them through a cation exchange column in the hydrogen form. They displace an equivalent amount of hydrogen ions that appear in the effluent and can be titrated. How many milli moles of potassium ion are contained in a liter of solution if the effluent obtained from a 5.00 ml aliquot run through a cation exchange required 26.7 ml of 0.0506 M Noah for titration?	05
2	**250**	The concentrations of Cl^-, NO_3^-, and SO_4^{2-} may be determined by ion chromatography. A 50-mL standard sample of 10.0-ppm Cl^-, 2.00-ppm NO_3^-, and 5.00-ppm SO_4^{2-} gave signals (in arbitrary units) of 59.3, 16.1, and 6.08, respectively. A sample of effluent from a wastewater treatment plant was diluted tenfold, and a 50-mL portion gave signals of 44.2 for Cl^-, 2.73 for NO_3^-, and 5.04 for SO_4^{2-}. Report the parts per million of each anion in the effluent sample.	05
2	**251**	The total cation content of natural water is often determined by exchanging the cations by hydrogen ions on a strong acid ion exchange resin. A 25.0 ml sample of natural water was diluted to 100.0 ml with distilled water and 2.0 g of cation exchange resin were added. After stirring, the mixture was filtered, and the solid remaining on filter paper was washed with three 15.0 ml portions of water. The filtrate and washings required 15.3 ml of 0.0202 M NaOH to give a bromocresol green end point. Calculate the number of milliequivalents of cation present in exactly 1.00 L of sample (here the equivalent weight of cation is its formula weight divided by its charge)	05
2	**252**	An aqueous solution containing $MgCl_2$ and HCl was analysed by first titrating a 25.00 ml aliquot to a bromocresol green end point with 18.96 ml of 0.02762 M NaOH. A 10.0 ml aliquot was then diluted to 50.0 ml with distilled water and passed through a strong acid ion exchange resin. The	05

		eluate and washings required 36.54 ml of the NaOH solution to reach the same end point. Report the molar concentration of HCl and $MgCl_2$ in the sample	
2	**253**	The method takes advantage of the reaction of water with calcium carbide, CaC_2, to produce acetylene gas, C_2H_2. By carrying out the reaction in a sealed vial, the amount of acetylene produced may be determined by sampling the headspace. In a typical analysis a sample of soil is placed in a sealed vial with CaC_2. Analysis of the headspace gave a blank-corrected signal of 2.70×10^5. A second sample is prepared in the same manner except that a standard addition of 5.0 mg H_2O/g solid is added, giving a blank-corrected signal of 1.06×10^6. Determine the number of milligrams of H_2O/g soil in the soil sample.	05
2	**254**	An aqueous solution containing $MgCl_2$ and HCl was analysed by first titrating a 25.00 ml aliquot to a bromocresol green end point with 19 ml of 0.03 M NaOH. A 10.0 ml aliquot was then diluted to 50.0 ml with distilled water and passed through a strong acid ion exchange resin. The eluate and washings required 35 ml of the NaOH solution to reach the same end point. Report the molar concentration of HCl and $MgCl_2$ in the sample.	05
2	**255**	An organic acid was isolated and purified by recrystallization of its barium salt. To determine the equivalent weight of the acid, a 0.393 g sample of salt was dissolved in about 100 ml of water. The solution was passed through a strong acid ion exchange resin and the column was then washed with water; the eluate and washing were titrated with 18.0 ml of 0.1006 M NaOH to a phenolphthalein end point. Calculate the equivalent weight of organic acid	05
2	**256**	An organic acid was isolated and purified by recrystallization of its barium salt. To determine the equivalent weight of the acid, a 0.393 g sample of salt was dissolved in about 100 ml of water. The solution was passed through a strong acid ion exchange resin and the column was then washed with water; the eluate and washing were titrated with 18.0 ml of 0.1006 M NaOH to a phenolphthalein end point. A potentiometric titration curve of the solution resulting when a second sample was treated in the same way revealed two end points: one at pH 5.0 and other at pH 9.0 what is the molecular weight of the acid?	05
3	**257**	Discuss the factors influencing the mobility of a species in electrophoresis.	05
3	**258**	What are the factors that are responsible for the separation of bio-components by electrophoresis? Explain the theory of electrophoresis with reference to Electrophoretic and electrosmotic mobility	05

3	**259**	Explain the principle of capillary electrochromatography. Mention its advantages.	05
3	**260**	Explain the basic principle and operation of electrophoresis. What are its applications ?	05
3	**261**	Discuss the factors influencing the mobility of a species in electrophoresis.	05
3	**262**	With a neat labelled diagram explain the instrumentation and process involved in the capillary electrophoresis. Discuss the controlling of joule heating technique.	05
3	**263**	Describe the basis of separation components in the electrophoresis technique.	05
3	**264**	Explain the principle of capillary zone electrophoresis for the separation of the components	05
3	**265**	What is the principle of micellar electrokinetic capillary chromatography? How does it differ from CZE.	05
3	**266**	Describe a major advantage electrokinetic capillary chromatography over conventional chromatography	05
3	**267**	Distinguish electrophoretic mobility and electroosmotic flow in capillary electrophoresis	05
3	**268**	Explain capillary zone electrophoresis and micellar electrokinetic capillary chromatography	05
3	**269**	Explain the separation of neutral components by micellar electrokinetic chromatography	05
3	**270**	Describe briefly the separation of uncharged components by micellar electrokinetic chromatography	05
3	**271**	What is the principle of CZE? How does it differ from micellar electrokinetic capillary chromatography.	05
3	**272**	List the various factors affecting solvent extraction. Explain any one in detail	05
3	**273**	Discuss the effect of pH and reagent concentration on the solvent extraction of metal chelates with suitable mathematical derivation	05
3	**274**	Describe two principal solvent extraction systems for metal ions. Give examples of each.	05
3	**275**	Discuss the effect of the pH and of the reagent concentration on the solvent extraction of metal chelates	05
3	**276**	Extraction of carboxylic acid is a pH dependent. Justify with suitable mathematical relation.	05
3	**277**	Extraction of iodine is dependent on iodide concentration, use mathematical equations to justify the statement	05
3	**278**	What are the two basic approaches adopted to achieve selectivity in the metal	05

		ion extraction by chelating agents? Why do we generally prefer an alkylphosphorus acid for the extraction of a metal ion over a carboxylic acid?	
3	**279**	The extraction of uranyl nitrate in a solvating extractant is enhanced by the addition of equal amounts of LiNO3, NaNO3 and CsNO3 to the aqueous phase. Arrange these alkali metal nitrates in their capability to enhance the extraction. Give reasons also.	05
3	**280**	Generally, what is the order of extraction power of high molecular weight amines? Give a possible reason if a deviation in the sequence is observed. What is the common reason for the selectivity in the extraction by crown ethers?	05
3	**281**	A certain metal complex, MXn.2H2O, formed with the chelating agent HX is poorly extracted in a nonpolar solvent but its extraction is favoured in a polar solvent. Explain why? What are the two basic approaches adopted to achieve selectivity in the metal ion extraction by chelating agents?	05
3	**282**	Highlight the differences between chelating agents and acidic organophosphorus esters, carboxylic acids as metal ion extractants.	05
3	**283**	High molecular weight amines extract the metal ions by a mechanism which is similar to their uptake by solid anion exchangers. In what respects, extraction in high molecular weight amines scores over sorption of metal ions by solid anion exchangers?	05
3	**284**	What is synergic extraction? Indicate its benefits in the extraction of metals as neutral chelates	05
3	**285**	Explain the batch and continuous extraction systems. List their applications.	05
3	**286**	Arrange the following three systems in increasing order of extraction: propionic acid – tributyl amine,valeric acid – tributyl amine, valeric acid – pyridine. Justify the statement	05
3	**287**	Discuss in detail the main role of a masking agent in the extraction equilibria of a metal chelate	05
3	**288**	What will be the problem with the extraction system if the density difference of the contacting phases is low and the interfacial tension of the organic phase is also low?	05
3	**289**	What will it result to if the extractant has poor metal loading capacity and regeneration power?	05

3	**290**	In the extraction of Cd(II) from oxalic acid medium in a high molecular weight quaternary ammonium oxalate the log- log plot of D vs free amine concentration gave a straight line of slope two. Propose the extraction equilibria.	05
3	**291**	Generally, the extraction efficiency of the various HMWA follow the sequence primary < secondary < tertiary < quaternary. In what situations a deviation in the above sequence is generally observed?	05
3	**292**	How does solid phase extraction differs from solvent extraction?	05
3	**293**	Bring out the difference between the solid phase extraction and solvent extraction.	05
3	**294**	Write the principles of solid phase extraction, and explain the sorbents used in solid phase extraction.	05
3	**295**	What is solid phase micro extraction? How does solid phase extraction differ from solvent extraction	05
3	**296**	For a solute with a distribution ratio of 25.0 show by calculation which is more effective, extraction of 10.0 ml of an aqueous solution with 10 ml organic solvent or extraction with two separate 5.0 ml portions of organic solvent.	05
3	**297**	Justify the statement "the best results are obtained with a relatively large number of extractions with small amount of solvent"	05
3	**298**	The caffeine in a sample of tea is extracted by combining 80.0 ml of tea with 15.0 ml of dichloromethane. If 89% of the caffeine is extracted in single step, what is the distribution ratio of caffeine under these conditions?	05
3	**299**	One gram of solute is contained in 100 ml of an aqueous solution. Calculate the amonut remaining in the aqueous phase after: a single extraction with 90 ml of organic solvent with appropriate reagents: and three successive extraction with 30 ml solvent. Comment on the results. Given the distribution coefficient between organic solvent and water is 90	05
3	**300**	The analysis of NO_3^- in aquarium water was carried out by CZE using IO_4^- as an internal standard. A standard solution of 15.0-ppm NO_3^- and 10.0-ppm IO_4^- gives peak heights (arbitrary units) of 95.0 and 100.1, respectively. A sample of water from an aquarium is diluted 1:100, and sufficient internal standard added to make its concentration 10.0 ppm. Analysis gives signals of	05

		29.2 and 105.8 for NO_3^- and IO_4^-, respectively. Report the parts per million of NO_3^- in the sample of aquarium water.	
3	**301**	What is the minimum distribution co-efficient that permits removal of 99% of a solute from 50.0 ml of water with a) two 25.0 ml extractions with toluene b) five 10.0 ml extractions with toluene?	05
3	**302**	For a solute with a distribution ratio of 25.0 show by calculation which is more effective, extraction of 10.0 ml of an aqueous solution with 10 ml organic solvent or extraction with two separate 5.0 ml portions of organic solvent.	05
3	**303**	The distribution constant for X between n-hexane and water is 9.6. Calculate the concentration of X remaining in the aqueous phase after 50.0 ml of 0.150 M 'X' are treated by extraction with following quantities of n-hexane: (a) one 40.0 ml portion (b) two 20.0 ml portion	05
3	**304**	The distribution coefficient for Z between n-hexane and water is 6.25. Calculate the percent of Z remaining in 25.0 ml of water that was originally 0.0600 M in Z after extraction with following volumes in n-hexane (a) one 25.0 ml portion (b) two 12.5 ml portion	05

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Question Paper Pattern- Model Question Paper

Q.P Code: 55101

St. Philomena's College (Autonomous) Mysore
II Semester M.Sc. Final Examination : May - 2019
Subject: ANALYTICAL CHEMISTRY
Title: SEPARATION TECHNIQUES

Time: 3 Hours **Max Marks: 70**

PART –A

Answer the following questions: **10×2=20**

1. a. Differentiate between isocratic and gradient elusion.
   b. List out the criteria for the selection of mobile and stationary phase in HPLC.
   c. Give the qualitative and quantitative applications of column chromatography
   d. Write the requirements of ion-exchange resin.
   e. Briefly discuss the properties of a supercritical fluid.
   f. What do you mean by "matrix activation" in affinity chromatography?
   g. List out the applications of thin layer chromatographic techniques.
   h. Distribution co-efficient is a pH dependent. Justify.
   i. In solvent extraction of uranium with 8 – hydroxy quinoline in chloroform the volumes of the aqueous and organic phase wire 25 *ml* when the parentage of extraction was 99.8%. Calculate the distribution ratio.
   j. Write the principles of solid phase extraction.

PART –B

Answer any FIVE of the following: **5×10=50**

2. a. How chromatographic separation is optimized using capacity factor, column selectivity and column efficiency.
   b. With neat diagram explain the working of FID in GC. **6+4**
3. a. Write a note on detectors used in HPLC and Applications of HPLC.
   b. The substance *x* and *y* had retention times of 16.4 and 17.63 minutes on a 30 *cm* column. Unretained species had 1.30 min. retention time. The peak width (W) was 1.11 and 1.21 min. respectively. Calculate: column resolution, average number of plates, plate height and length of the column required to achieve 1.5 resolutions. **5+5**

PTO

4 a. How anionic and cationic resins are synthesized?

b. Write a short note on apparatus, methodology and applications of TLC. **4+6**

5 a. How separations are achieved in affinity chromatography? What is the role of spaces arms? Write the applications of affinity chromatography.

b. With neat diagram explain mechanism of working and operating variables of SFC. What are the advantages of SFC over the other types of chromatography? **5+5**

6. a. Discuss types of extraction systems.

b. Show that $\log D = n(P^H - P^{H\,1/2})$ for extraction of uncharged metal chelates. **4+6**

7. a. What do you understand by:

i) Capillary zone electrophoresis

ii) Micellar kinetic capillary chromatography

iii) Capillary gel electrophoresis

b. Calculate the *mg* of iron (III) left unextracted from 100 *ml* of solution containing 200 *mg* of iron (III) in 6 *m HCl* after (i) one extraction with 100 *mL* (ii) two extractions with 50 *ml* each and (iii) four extractions with 25 *ml* portions of ethyl ether. The distribution ratio for the extraction is 150. **6+4**

8. a. Give the principle and experimental methodology involved in gel filtration chromatography.

b. Write the principle and explain the instrumentation of Gas chromatography. **4+6**

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